

Colby Miller

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EDUCATION

Bachelor of Science in Computer Engineering

University of Utah

Expected May 2029

Salt Lake City, UT

Relevant coursework: Circuits I (AC/DC Analysis), Physics for Scientists & Engineers II (E&M), Calculus I & II, Differential Equations & Linear Algebra, Object-Oriented Programming (Java), Data Structures & Algorithms

High School Diploma with Honors • 3.98 GPA • National Honor Society

Brighton High School

2023

Cottonwood Heights, UT

TECHNICAL SKILLS

HDL & verification: Verilog (RTL), self-checking testbenches, golden-model comparison, mutation testing

Digital design: Combinational & sequential logic, SPI, direct digital synthesis (DDS), sigma-delta modulation

FPGA & ASIC flow: Lattice iCE40UP5K (iCEBreaker), Yosys, nextpnr, TinyTapeout (SkyWater SKY130)

Simulation & formal: Icarus Verilog, GTKWave; BLIF, ABC, SAT solvers (research lab)

Programming: Java (OOP, data structures), Python (coursework), Git/GitHub

Hardware & IT: PC assembly, UEFI/BIOS setup, hardware/software fault isolation, Chromebook/printer repair

Languages: Spanish (professional working proficiency)

PROJECTS & SELF-DIRECTED WORK

ASIC Tapeout: 3-Voice Polyphonic Synthesizer (TinyTapeout)

Jun 2026 - Aug 2026

- Designed a 3-voice synthesizer in Verilog (SPI control, DDS core, sigma-delta audio out); prototyped on an iCEBreaker FPGA and taped out on TinyTapeout's SKY26c shuttle (SkyWater 130nm).
- Verified the design with directed testbenches, golden-model comparison, and mutation testing, then confirmed it in hardware by playing music through headphones on the FPGA prototype before tapeout.

PC Building & Systems Integration

2020 - Present

- Built 12+ custom PCs for friends, family, and personal use, including one built for resale and sold through KSL Classifieds.
- Selected compatible components across a range of budgets; diagnosed and resolved issues at the hardware, UEFI/BIOS, and software levels.

EXPERIENCE

Production Professional (Back of House)

Jul 2026 - Present

Chick-fil-A

Murray, UT

- Produce ~250 lb of fries per 4-hour peak shift, keeping pace with ~70 orders per hour.
- Prepare food to standardized recipes and food-safety procedures as part of a fast-paced back-of-house team.

Full-Time Missionary (Spanish-Speaking)

Jun 2024 - Jun 2026

The Church of Jesus Christ of Latter-day Saints, Texas Houston East Mission

Houston, TX

- Taught ~30 lessons per week in Spanish, explaining complex ideas clearly and adapting to each learner.
- Scored above the fluency benchmark on a standardized Spanish speaking assessment.
- Trained two new missionaries; managed daily schedules and priorities with minimal supervision.

Engineering Scholars Program: Formal Verification Research

Fall 2023 - Spring 2024

University of Utah, Prof. Priyank Kalla's lab

Salt Lake City, UT

- Built digital logic foundations from no prior background as a first-year participant in research on polynomial-based formal verification methods.
- Set up gate-level circuit files in BLIF format for verification experiments using ABC and SAT solvers.

IT Intern

Sep 2022 - Jul 2023

Canyons School District

Sandy, UT

- Diagnosed, triaged, and repaired hundreds of Chromebooks across multiple sites, isolating hardware/software faults.
- Resolved 5-10 tickets per day; handled ~15 help desk calls per day during the summer.
- Serviced onboard computers and camera systems on school buses; maintained printers.